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SECTION : G	SUB: SEP

DISASTER RELIEF SUPPLY & VOLUNTEER COORDINATOR

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall match incoming field shelter supply requisitions against available central warehouse supplies and allocate delivery priority.	**High**	**Pass:** Critical need is given the highest dispatch priority and routed to the logistics manifest. **Fail:** A low-priority item is dispatched ahead of water or medical needs.	Ensures scarce relief supplies reach the most urgent needs first.	Given requirement.
FR-002	Functional	The system shall allow a Relief Volunteer to submit a shelter supply need report specifying shelter location, item category, and quantity.	**High**	A submitted report containing location, category, and quantity appears in the Disaster Manager's queue within **5 seconds**.	Field reporting is the entry point for downstream matching and dispatch.	Reports must be geotagged.
FR-003	Functional	The system shall dispatch a Relief Volunteer to a delivery task only if their registered skill	**High**	Only volunteers tagged with the required skill appear as eligible. Assigning a non-matching volunteer	Prevents volunteers from being assigned to tasks for which they	Skill tags must be pre-verified.

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		set matches the task's required skill(s).		is blocked and an error message is displayed.	are not qualified.	
FR-004	Functional	The system shall record and update the real-time delivery status (Dispatched, In-Transit, Delivered, Delayed) of each relief vehicle.	**Medium**	A status change made by a volunteer or GPS update is reflected on the dashboard within **10 seconds**.	Enables the Disaster Manager to monitor delivery progress and identify stalled shipments.	Status updates should support manual and GPS input.
FR-005	Functional	The system shall allow a Disaster Manager to view a consolidated dashboard of open supply requests, active dispatches, and delivery statuses.	**Medium**	All open requests, active dispatches, and current statuses are displayed and match the underlying database within **2 seconds**.	Provides the Disaster Manager with a single view for coordinating relief operations.	Dashboard should support offline caching.
NFR-001	Non-Functional	The disaster mapping dashboard must operate with low network bandwidth consumption and support offline GIS data caching.	**High**	**Pass:** Benchmarking tests confirm that the dashboard meets the defined latency target and remains usable under simulated low-bandwidth conditions. **Fail:** The dashboard becomes unusable or cached GIS data	Disaster areas may have poor or no network connectivity.	Given requirement.

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				cannot be accessed offline.		
NFR-002	Non-Functional	The system shall preserve all shelter supply requests and dispatch actions entered during a network outage and automatically sync them to the central server within **30 seconds** of reconnection, without data loss.	**High**	In a simulated **5-minute network outage**, 100% of locally entered requests and dispatch actions shall sync correctly within **30 seconds** of reconnection, with no duplicate or missing records.	Prevents loss of critical relief data during unreliable field connectivity.	Sync conflicts should be detected and resolved without data loss.

Actors

1. Relief Volunteer
2. Disaster Manager
3. Warehouse System (external)

Main Use Cases

1. Report Shelter Supply Need
2. Match and Allocate Supplies
3. Verify Warehouse Inventory
4. Dispatch Volunteer
5. Verify Volunteer Skill Match

6. Track Relief Vehicle Delivery
7. Send Delay Alert
8. View Disaster Dashboard

Important UML Relationships

- Match and Allocate Supplies «include» Verify Warehouse Inventory
- Dispatch Volunteer «include» Verify Volunteer Skill Match
- Send Delay Alert «extend» Track Relief Vehicle Delivery